

CASA0029: Urban Data Visualisation

Open Source Mapping Practical

This practical provides examples of open source interactive mapping solutions for large spatial data files, using MapLibre JS, Leaflet JS and Protomap Tiles (PMTiles). This is an alternative solution to the main mapping library we have been using, Mapbox, which is no longer an open source solution.

MapLibre: an Open Source Alternative Web Mapping Library

When Mapbox switched to a commercial license, MapLibre launched as an open source alternative based on the last fully open source version of Mapbox. MapLibre GL JS is a very similar library to Mapbox GL JS, with the same functionality and syntax-

<https://maplibre.org/>

<https://maplibre.org/maplibre-gl-js/docs/examples/>

We have also been using Mapbox to host our spatial datasets. If we switch to MapLibre then we no longer have access to the Tileset functionality where Mapbox hosted our spatial data as scalable vector tiles. We therefore need an alternative hosting solution, and we are going to use PMTiles.

Protomap Tiles (PMTiles): A Scalable Flat File Solution for Large Spatial Datasets

Geospatial technologists have been working on solutions to host large spatial datasets on standard web space without the need for a full database server. An innovative solution has come from Protomaps-

<https://docs.protomaps.com/>

<https://docs.protomaps.com/pmtiles/>

PMTiles is an archive of spatial data tiles (either vector or raster tiles) similar to the Mapbox MBTiles format. Unlike MBTiles however, PMTiles is set up to use HTTP Range Requests to serve spatial data on standard web hosting in a scalable and performant manner. You can view a good demo of PMTiles mapping OpenStreetMap data of the whole world (125gb) on their demo site-

<https://maps.protomaps.com/>

Example 1: Displaying a Vector PMTiles File in MapLibre

In our first example, we will display a simple PMTiles file of World Borders using MapLibre.

Download the “OpenSourceMappingDataExamples.zip” file from Moodle and unzip this. You will see a “WorldBorders.pmtiles” file, which was created from a GeoJSON file using [Tippecanoe](#) (we will discuss creating PMTiles files more below).

To view the PMTiles file we use MapLibre. In the example below, you can see that we first load the MapLibre library, then we load an additional PMTiles library in the header. The PMTiles protocol also needs to be instantiated in the main script. The creation of the MapLibre map uses the same syntax as Mapbox, with the data source stating ‘pmtiles://’ plus the URL of the PMTiles file we are using-

MapLibrePMTiles1_VectorWorldBorders.html

```
<!DOCTYPE html>
<html lang="en">
<head>
  <title>PMTiles source and protocol example using MapLibre</title>
  <meta name="viewport" content="width=device-width, initial-scale=1">
  <link rel="stylesheet" href="https://unpkg.com/maplibre-gl@5.18.0/dist/maplibre-gl.css" />
  <script src="https://unpkg.com/maplibre-gl@5.18.0/dist/maplibre-gl.js"></script>
  <script src="https://unpkg.com/pmtiles@3.2.0/dist/pmtiles.js"></script>
  <style>
    body { margin: 0; padding: 0; }
    html, body, #map { height: 100%; }
  </style>
</head>
<body>
<div id="map"></div>
<script>

  // add the PMTiles plugin to the maplibregl global.
  const protocol = new pmtiles.Protocol();
  maplibregl.addProtocol('pmtiles', protocol.tile);

  const PMTILES_URL = 'WorldBorders.pmtiles';

  const map = new maplibregl.Map({
    container: 'map',
    zoom: 1,
    center: [0,0],
    style: {
      version: 8,
      sources: {
        'example_source': {
          type: 'vector',
          url: `pmtiles://${PMTILES_URL}`
        }
      },
      layers: [
        {
          'id': 'borders',
          'source': 'example_source',
          'source-layer': 'WorldBorders',
          'type': 'fill',
          'paint': {
            'fill-color': 'steelblue',
            'fill-outline-color': '#444',
          }
        }
      ]
    }
  });

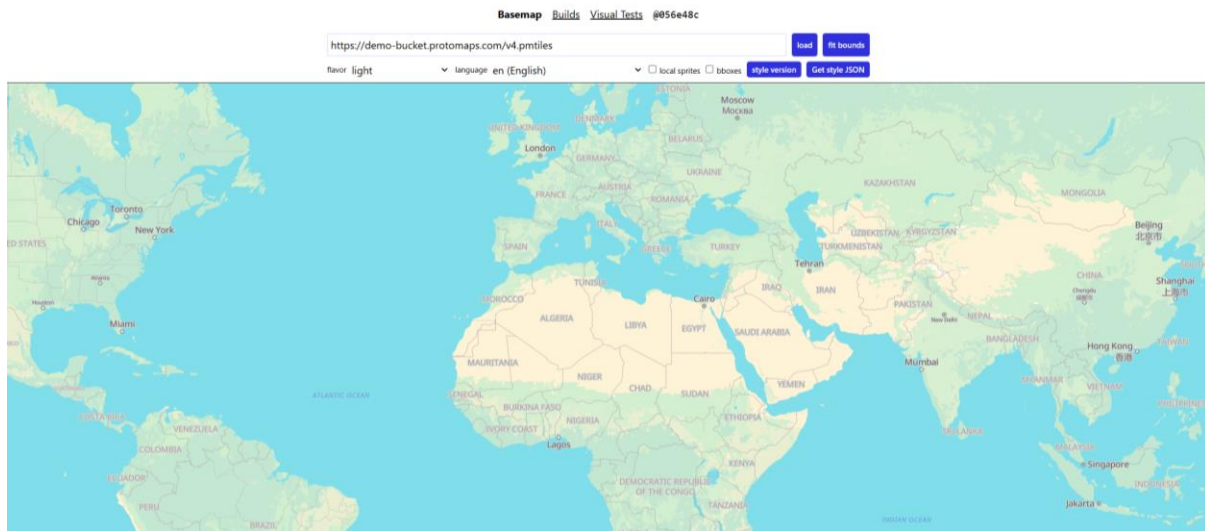
  map.addControl(new maplibregl.NavigationControl({
    visualizePitch: false,
    visualizeRoll: false,
    showZoom: true,
    showCompass: true
  }));

</script>
</body>
</html>
```

Displaying OpenStreetMap Data Using PMTiles

Protomaps offer a number of tools for using PMTiles with OpenStreetMap data. Their demo site has builds of global OSM data to use as basemaps-

<https://maps.protomaps.com/builds/>



Rather than use the entire OSM dataset (125gb) we can use the PMTiles Command Line tool to extract a subset of the global file-

<https://docs.protomaps.com/guide/getting-started>

<https://docs.protomaps.com/pmtiles/cli#extract>

You can extract a spatial subset of OSM data using a bounding box, or a GeoJSON defining a region of interest. In this case we are going to limit the file by zoom level using the following command-

pmtiles extract https://build.protomaps.com/20260301.pmtiles planet_z7.pmtiles --maxzoom=7

This has already been done for you in the "OSMglobal_z7.pmtiles" file, which is a more manageable 180mb.

Example 2: OSM PMTiles Map Using MapLibre

The code to visualise the OSM data in MapLibre is very similar to our previous example. Rather than hand-coding the style, this time we are using the standard Style JSON that is defined on the Protomaps website (use the 'Get Style JSON' button)-

<https://maps.protomaps.com/#flavorName=white&lang=en&map=0.8/0/0>

Designing styles for large topographic datasets at multiple zoom levels can take a long time, and it is useful to have standard styles available. Note that Protomaps provide several options, and you can swap the "OSMLightStyle.json" file for the "OSMDataVizWhiteStyle.json" file in the example below to see the change in OSM appearance using these styles.

MapLibrePMTiles2_OpenStreetMap.html

```
<!DOCTYPE html>
<html lang="en">
<head>
  <title>PMTiles Example using OpenStreetMap Data</title>
```

```

<meta property="og:description" content="Uses the PMTiles plugin and protocol to
present a map." />
<meta charset='utf-8'>
<meta name="viewport" content="width=device-width, initial-scale=1">
<link rel='stylesheet' href='https://unpkg.com/maplibre-gl@5.18.0/dist/maplibre-
gl.css' />
<script src='https://unpkg.com/maplibre-gl@5.18.0/dist/maplibre-gl.js'></script>
<script src="https://unpkg.com/pmtiles@3.2.0/dist/pmtiles.js"></script>
<style>
  body { margin: 0; padding: 0; }
  html, body, #map { height: 100%; }
</style>
</head>
<body>
<div id="map"></div>
<script>

  // add the PMTiles plugin to the maplibregl global.
  const protocol = new pmtiles.Protocol();
  maplibregl.addProtocol('pmtiles', protocol.tile);

  const PMTILES_URL = 'OSMglobal_z7.pmtiles';

  const map = new maplibregl.Map({
    container: 'map',
    zoom: 2,
    center: [0,0],
    style: "OSMLightStyle.json" // You can get standard basemap Style JSON files
from https://maps.protomaps.com/ Try swapping style for OSMDataVizWhiteStyle.json
  });

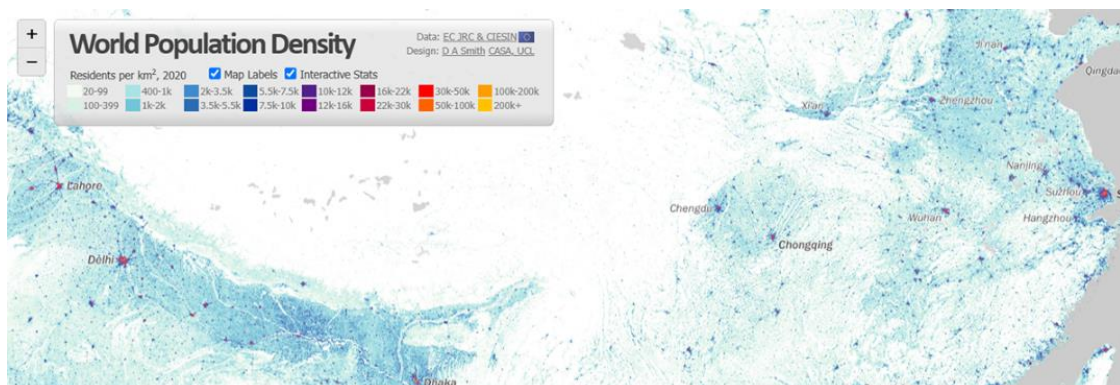
  map.addControl(new maplibregl.NavigationControl({
    visualizePitch: false,
    visualizeRoll: false,
    showZoom: true,
    showCompass: true
  }));

</script>
</body>
</html>

```

Raster PMTiles File Example

The PMTiles format will also work for hosting raster tiles as well. PMTiles have the advantage of higher rates of compression for raster files, and can also containing the thousands of tile files as a single archive. The example below visualises global population density data from a PMTiles file, this time using the Leaflet library.



Again we need to load a PMTiles plugin to be able to use the PMTiles file “WorldPopDen2023_z0-7.pmtiles”-

LeafletPMTiles1_RasterWorldPopDen.html

```
<html>
  <head>
    <title>PMTiles Leaflet Example</title>
    <meta charset="utf-8"/>
    <link rel="stylesheet" href="https://unpkg.com/leaflet@1.9.0/dist/leaflet.css"
  />

    <script src="https://unpkg.com/leaflet@1.9.0/dist/leaflet.js"></script>
    <script src="https://unpkg.com/pmtiles@4.4.0/dist/pmtiles.js"></script>
    <style>
      body, #map {
        height:100vh;
        margin: 0;
      }
    </style>
  </head>
  <body>
    <div id="map"></div>
    <script type="text/javascript">

      const mymap = L.map('map').setView([51.505, -0.09], 5);

      const p = new pmtiles.PMTiles(
        "WorldPopDen2023_z0-7.pmtiles"
      );

      p.getHeader().then((h) => {
        let layer = pmtiles.leafletRasterLayer(p, {
          maxNativeZoom: 8
        });
        layer.addTo(mymap);
      });

    </script>
  </body>
</html>
```

Creating PMTiles Files: Vector Data

The Protomaps website provides a useful guide on creating vector PMTiles files

<https://docs.protomaps.com/pmtiles/create>

[Tippecanoe](#) is the best tool for converting your own spatial data to PMTiles format. It uses input data in GeoJSON format, and is particularly good for scaling vector data across zoom levels for high quality web mapping. Unfortunately, this tool is only available for MacOS, though you can use Docker as an alternative way of running this software on Windows.

If your data is in MBTiles format (the Mapbox vector tiles format) then you can convert this to PMTiles using the PMTiles Command Line Tool- <https://docs.protomaps.com/pmtiles/create>

For example, this can be used to convert the Ordnance Survey OpenZoomStack data (which can be downloaded as MBTiles from Digimap, including great Style JSON files) to PMTiles-

https://luminocity3d.org/CASATest/MapLibrePMTiles3_OSZoomStackOutdoors.html

Creating PMTiles Files: Raster Data

For raster data, you can use QGIS to create XYZ files and MBTiles, and then convert these outputs to PMTiles. Note for XYZ directories, you may need the additional step of converting this to MBTiles using MB-Util. Using QGIS means you can design the raster styling using the QGIS interface and then export this-

https://docs.qgis.org/3.40/en/docs/user_manual/processing_algs/qgis/rastertools.html#generate-xyz-tiles-directory

https://docs.qgis.org/3.40/en/docs/user_manual/processing_algs/qgis/rastertools.html#generate-xyz-tiles-mbtiles

This approach generates static tiles, and is not appropriate for dynamic data that changes frequently. But for more static data, this method can be effective. The QGIS export route was used on the World Population Density map website- <https://luminocity3d.org/WorldPopDen/>

Further Materials on MapLibre and PMTiles

MapLibre is being actively developed and you can see more examples and future plans at the following sites-

<https://maplibre.org/>

<https://maplibre.org/maplibre-gl-js/docs/API/>

<https://maplibre.org/maplibre-gl-js/docs/examples/>

<https://maplibre.org/roadmap/>

It is also interesting that MapLibre has recently developed a new vector tiles format similar to PMTiles, called MLTiles. This launched in early 2026 and is not yet widely supported, but may become a popular format in the future-

<https://maplibre.org/news/2026-01-23-mlt-release/>

If you are creating your own PMTiles files, then the PMTiles Viewer is a useful site for quick testing-

<https://pmtiles.io/>